Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The values of α for which the vector $\vec{a} = \hat{i} + 3\hat{j} + (\sin 2\alpha)\hat{k}$
makes an obtuse angle with the z-axis and the vectors $\overrightarrow{\mathrm{b}} = (\tan\alpha)\hat{\mathrm{i}} - \hat{\mathrm{j}} + 2\sqrt{\sin\frac{\alpha}{2}}\hat{\mathrm{k}}$
and $\overrightarrow{\mathrm{c}} = (\tan\alpha)\hat{\mathrm{i}} + (\tan\alpha)\hat{\mathrm{j}} - 3\sqrt{\operatorname{cosec}\frac{\alpha}{2}}\hat{\mathrm{k}}$
are mutually orthogonal are

(A) $\mathrm{n}\pi - \tan^{-1} 2, \mathrm{n} \in \mathbf{I}$

(B) $(4n+2)\pi - \tan^{-1} 2, n \in \mathbf{I}$

(C) $2n\pi - \tan^{-1} 2, n \in \mathbf{I}$

(D) $(2\mathrm{n}+1)\pi - \tan^{-1} 2, \mathrm{n} \in \mathbf{I}$

Q.2

If the pth , q^{th} and r^{th} terms of a G.P. are positive numbers a, and c respectively, then find the angle between the vectors $\log a^2\hat{i} + \log b^2\hat{j} + \log c^2\hat{k}$ and
$(\mathrm{q}-\mathrm{r})\hat{\mathrm{i}} + (\mathrm{r}-\mathrm{p})\hat{\mathrm{j}} + (\mathrm{p}-\mathrm{q})\hat{\mathrm{k}}$

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{2}$

Q.3

Force $\hat{i} + 2\hat{j} - 3\hat{k}, 2\hat{i} + 3\hat{j} + 4\hat{k}$ and $-\hat{i} - \hat{j} + \hat{k}$ are acting at
the point $\mathrm{P}(0,1,2)$. The moment of these forces about the point $A(1,-2,0)$ is

(A) $2\hat{i} - 6\hat{j} + 10\hat{k}$

(B) $-2\hat{i} + 6\hat{j} - 10\hat{k}$

(C) $2\hat{i} + 6\hat{j} - 10\hat{k}$

(D) None of these

Q.4

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If $\vec{a} = \hat{i} + \hat{j}, \hat{b} = 2\hat{j} - \hat{k}$ and $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$ $\overrightarrow{\mathrm{r}} \times \overrightarrow{\mathrm{b}} = \overrightarrow{\mathrm{a}} \times \overrightarrow{\mathrm{b}}$, then what is the value of $\frac{\overrightarrow{\mathrm{r}}}{|\overrightarrow{\mathrm{r}}|}$?

(A) $\frac{(\hat{\mathrm{i}}+3\hat{\mathrm{j}}-\hat{\mathrm{k}})}{\sqrt{11}}$

(B) $\frac{(\hat{\mathrm{i}}-3\hat{\mathrm{j}}+\hat{\mathrm{k}})}{\sqrt{11}}$

(C) $\frac{(\hat{\mathrm{i}}+3\hat{\mathrm{j}}+\hat{\mathrm{k}})}{\sqrt{11}}$

(D) $\frac{(\hat{\mathrm{i}}-3\hat{\mathrm{j}}-\hat{\mathrm{k}})}{\sqrt{11}}$

Q.5

Let $\overrightarrow{\mathrm{A}} = 2\hat{\mathrm{i}} + \hat{\mathrm{k}}, \overrightarrow{\mathrm{B}} = \hat{\mathrm{i}} + \hat{\mathrm{j}} + \hat{\mathrm{k}}$ and $\overrightarrow{\mathrm{C}} = 4\hat{\mathrm{i}} - 3\hat{\mathrm{j}} + 7\hat{\mathrm{k}}$. The vector $\overrightarrow{\mathrm{R}}$ which satisfies the equations $\overrightarrow{\mathrm{R}} \times \overrightarrow{\mathrm{B}} = \overrightarrow{\mathrm{C}} \times \overrightarrow{\mathrm{B}}$ and $\overrightarrow{\mathrm{R}} \cdot \overrightarrow{\mathrm{A}} = 0$ is given by

(A) $-2\hat{\mathrm{i}} + \hat{\mathrm{k}}$

(B) $-\hat{\mathrm{i}} - 8\hat{\mathrm{j}} + 2\hat{\mathrm{k}}$

(C) $\frac{1}{\sqrt{6}}(\hat{\mathrm{i}} - \hat{\mathrm{j}} + 2\hat{\mathrm{k}})$

(D) None of these

Q.6

What is the interior acute angle of the parallelogram whose sides are represented by the vectors $\frac{1}{\sqrt{2}}\hat{\mathrm{i}} + \frac{1}{\sqrt{2}}\hat{\mathrm{j}} + \hat{\mathrm{k}}$ and $\frac{1}{\sqrt{2}}\hat{\mathrm{i}} - \frac{1}{\sqrt{2}}\hat{\mathrm{j}} + \hat{\mathrm{k}}$?

(A) 60°

(B) 45°

(C) 30°

(D) 15°

Q.7

If three vectors $\overrightarrow{\mathrm{a}}, \overrightarrow{\mathrm{b}}, \overrightarrow{\mathrm{c}}$ are such that $\overrightarrow{\mathrm{a}} \neq 0$ and $\overrightarrow{\mathrm{a}} \times \overrightarrow{\mathrm{b}} = 2\overrightarrow{\mathrm{a}} \times \overrightarrow{\mathrm{c}}, |\overrightarrow{\mathrm{a}}| = |\overrightarrow{\mathrm{c}}| = 1, |\overrightarrow{\mathrm{b}}| = 4$ and the angle between $\overrightarrow{\mathrm{b}}$ and $\overrightarrow{\mathrm{c}}$ is $\cos^{-1}\frac{1}{4}$ then $\overrightarrow{\mathrm{b}} - 2\overrightarrow{\mathrm{c}} = \lambda\overrightarrow{\mathrm{a}}$ where λ is equal to

(A) ± 2

(B) ± 4

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(C) $\frac{1}{2}$

(D) $\frac{1}{4}$

Q.8

For any vector $\overrightarrow{p}$, the value of $\frac{3}{2}\left\{|\overrightarrow{p} \times \hat{i}|^2 + |\overrightarrow{p} \times \hat{j}|^2 + |\overrightarrow{p} \times \hat{k}|^2\right\}$ is

(A) $\overrightarrow{p}^2$

(B) $2\overrightarrow{p}^2$

(C) $3\overrightarrow{p}^2$

(D) $4\overrightarrow{p}^2$

Q.9

Let $A = x + 4y\vec{a} + 2x + y + 1\vec{b}$ and $B = y - 2x + 2\vec{a} + 2x - 3y - 1\vec{b}$ where $\vec{a}$ and $\vec{b}$ are non-collinear vectors, if $3A = 2B$ then

(A) $x = 1, y = 2$

(B) $x = 2, y = 1$

(C) $x = 2, y = -1$

(D) $x = -1,\ y = 2$

Q.10

What is a vector of unit length orthogonal to both the vectors $\hat{i} + \hat{j} + \hat{k}$ and $2\hat{i} + 3\hat{j} - \hat{k}$?

(A) $\frac{-4\hat{i}+3\hat{j}-\hat{k}}{\sqrt{26}}$

(B) $\frac{-4\hat{i}+3\hat{j}+\hat{k}}{\sqrt{26}}$

(C) $\frac{-3\hat{i}+2\hat{j}-\hat{k}}{\sqrt{14}}$

(D) $\frac{-3\hat{i}+2\hat{j}+\hat{k}}{\sqrt{14}}$

Q.11

If $\overrightarrow{OA} = \overrightarrow{a}; \overrightarrow{OB} = \overrightarrow{b}; \overrightarrow{OC} = 2\overrightarrow{a} + 3\overrightarrow{b}$
$\overrightarrow{OD} = \overrightarrow{a} - 2\overrightarrow{b}$, the length of $\overrightarrow{OA}$ is three times
the length of $\overrightarrow{OB}$ and $\overrightarrow{OA}$ is perpendicular to
$\overrightarrow{DB}$ then $(\overrightarrow{BD} \times \overrightarrow{AC}) \cdot (\overrightarrow{OD} \times \overrightarrow{OC})$ is

(A) $7|\overrightarrow{a} \times \overrightarrow{b}|^2$

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(B) $42|\vec{a} \times \vec{b}|^2$

(C) 0

(D) None of these

Q.12

Consider the parallelopiped with side $\overrightarrow{\mathrm{a}} = 3\hat{\mathrm{i}} + 2\hat{\mathrm{j}} + \hat{\mathrm{k}}$, $\overrightarrow{\mathrm{b}} = \hat{\mathrm{i}} + \hat{\mathrm{j}} + 2\hat{\mathrm{k}}$ and $\overrightarrow{\mathrm{c}} = \hat{\mathrm{i}} + 3\hat{\mathrm{j}} + 3\hat{\mathrm{k}}$ then the angle between $\vec{a}$ and the plane
containing the face determined by $\overrightarrow{\mathrm{b}}$ and $\overrightarrow{\mathrm{c}}$ is

(A) $\sin^{-1}\frac{1}{3}$

(B) $\cos^{-1}\frac{9}{14}$

(C) $\sin^{-1}\frac{9}{14}$

(D) $\sin^{-1}\frac{2}{3}$

Q.13

Statement I: Three points with position vectors $\vec{a},\vec{b},\vec{c}$ are collinear, if $\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}=0$

Statement II: $\overrightarrow{AB}\cdot\overrightarrow{AC}=0$, then $\overrightarrow{AB}\perp\overrightarrow{AC}$

(A) Statement I is true; Statement II is true; Statement II is the correct explanation of the statement I

(B) Statement $\mathrm{I}$is true; Statement II is true; Statement II is not the correct explanation of the Statement I

(C) Statement I is true; Statement II is false

(D) Statement I is false; Statement $\mathrm{II}$is true

Q.14

Let $x^2+3y^2=3$ be the equation of an ellipse in the $x-y$ plane. A and B are two points whose position vectors are $-\sqrt{3}\hat{i}$ and $-\sqrt{3}\hat{i}+2\hat{k}$. Then the position vector of a point P on the ellipse such that $\angle APB=\pi/4$ is

(A) $\pm\hat{j}$

(B) $\pm(\hat{i}+\hat{j})$

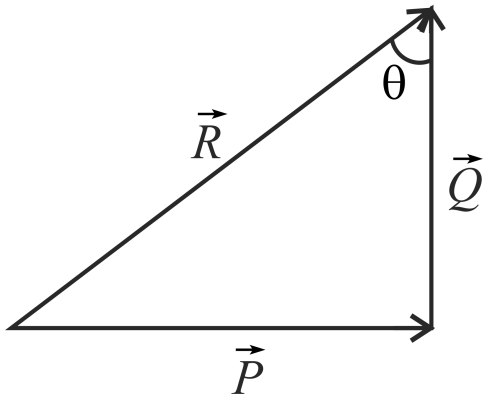
(C) $\pm\hat{i}$

(D) None of these

Q.15

If vectors $\vec{P}$, $\vec{Q}$ and $\vec{R}$ have magnitude 5, 12 and 13 units and $\vec{P}+\vec{Q}=\vec{R}$, then find the angle between $\vec{Q}$ and $\vec{R}$.

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(A) $\cos^{-1}\left(\frac{12}{13}\right)$

(B) $\cos^{-1}\left(\frac{11}{13}\right)$

(C) $\cos^{-1}\left(\frac{5}{13}\right)$

(D) $\cos^{-1}\left(\frac{1}{13}\right)$

Q.16

Given, $\vec{P} = \vec{A} + \vec{B}$ and $\vec{P} = \vec{A} + \vec{B}$. The angle between $\vec{A}$ and $\vec{B}$ is

(A) 0°

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{2}$

(D) π

Q.17

$\vec{a}, \vec{b}$ and $\vec{c}$ are three vectors with magnitude $|\vec{a}| = 4, |\vec{b}| = 4, |\vec{c}| = 2$ and such that $\vec{a}$ is perpendicular to $(\vec{b} + \vec{c}), \vec{b}$ is perpendicular to $(\vec{c} + \vec{a})$ and $\vec{c}$ is perpendicular to $(\vec{a} + \vec{b})$.
It follows that $|\vec{a} + \vec{b} + \vec{c}|$ is equal to:

(A) 9

(B) 6

(C) 5

(D) 4

Q.18

If vector $a = 2i - 3j + 6k$ and vector $b = -2i + 2j - k$, then $\frac{\text{Projection of vector } a \text{ on vector } b}{\text{Projection of vector } b \text{ on vector } a} =$

(A) $\frac{3}{7}$

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(B) $\frac{7}{3}$

(C) 3

(D) 7

Q.19

In the ΔOAB, M is the midpoint of AB, C is a point on OM, such that $2\overrightarrow{OC} = \overrightarrow{CM}$. X is a point on the side OB, such that $\overrightarrow{OX} = 2\overrightarrow{XB}$. The line XC is produced to meet OA at Y, then $\frac{OY}{YA}$ is equal to

(A) $\frac{1}{3}$

(B) $\frac{2}{7}$

(C) $\frac{3}{2}$

(D) $\frac{2}{5}$

Q.20

If $\mathrm{a}, \mathrm{b}, \mathrm{c}$ are the $\mathrm{p}^{\mathrm{th}}, \mathrm{q}^{\mathrm{th}} \cdot \mathrm{r}^{\mathrm{th}}$ terms of an HP and $\overrightarrow{\mathrm{u}} = (\mathrm{q}-\mathrm{r})\overrightarrow{\mathrm{i}} + (\mathrm{r}-\mathrm{p})\overrightarrow{\mathrm{j}} + (\mathrm{p}-\mathrm{q})\overrightarrow{\mathrm{k}}, \overrightarrow{\mathrm{v}} = \frac{\overrightarrow{\mathrm{i}}}{\mathrm{a}} + \frac{\overrightarrow{\mathrm{j}}}{\mathrm{b}} + \frac{\overrightarrow{\mathrm{k}}}{\mathrm{c}}$ then

(A) $\overrightarrow{\mathrm{u}}, \overrightarrow{\mathrm{v}}$ are parallel vectors

(B) $\overrightarrow{\mathrm{u}}, \overrightarrow{\mathrm{v}}$ are orthogonal vectors

(C) $\overrightarrow{\mathrm{u}} \cdot \overrightarrow{\mathrm{V}} = 1$

(D) $\overrightarrow{\mathrm{u}} \times \overrightarrow{\mathrm{v}} = \overrightarrow{\mathrm{i}} + \overrightarrow{\mathrm{j}} + \overrightarrow{\mathrm{k}}$

Answers & Solutions

Q.1 Answer:

$(2\mathrm{n}+1)\pi - \tan^{-1} 2, \mathrm{n} \in \mathbf{I}$

Solution:

$\overrightarrow{\mathrm{a}} \cdot \hat{\mathrm{k}} < 0 \Rightarrow \sin 2\alpha < 0$

$\overrightarrow{\mathrm{b}} \cdot \overrightarrow{\mathrm{c}} \Rightarrow \tan^2\alpha - \tan\alpha - 6 = 0 \Rightarrow \tan\alpha = -2$ **or 3**

$\sin 2\alpha < 0$ **if** $\tan\alpha = -2$

for which $\sin\frac{\alpha}{2} > 0$ **if** α **lies in the second quadrant** $\therefore \alpha = \pi + \tan^{-1}(-2) = \pi - \tan^{-1} 2$

General value is $\alpha = 2\mathrm{n}\pi + \pi - \tan^{-1} 2, \mathrm{n} \in \mathrm{I}$

Q.2 Answer:

$\frac{\pi}{2}$

Solution:

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Let A be the first term and x the common ratio of G.P.

So, $\mathrm{a} = \mathrm{Ax}^{\mathrm{p}-1} \Rightarrow \log \mathrm{a} = \log \mathrm{A} + (\mathrm{p}-1)\log \mathrm{x}$

Similarly, $\log \mathrm{b} = \log \mathrm{A} + (\mathrm{q}-1)\log \mathrm{x}$

and $\log c = \log A + (r-1)\log x$

If $\vec{\alpha} = \log a^2\hat{i} + \log b^2\hat{j} + \log c^2\hat{k}$

and $\vec{\beta} = (q-r)\hat{i} + (r-p)\hat{j} + (p-q)\hat{k}$ then

$\vec{\alpha}\cdot\vec{\beta} = 2[\log a(q-r) + \log b(r-p) + \log c(p-q)]$

$= 2[(q-r)\{\log A + (p-1)\log x\} + (r-p)\{\log A + (q-1)\log x\}$

$+(p-q)\{\log A + (r-1)\log x\}]$

$= 2[(q-r+r-p+p-q)\log A + (qp - pr - p + r + qr - pq$

$-r + p + pr - qr - p + q)\log x] = 0$

Hence, the angle between $\vec{\alpha}$ and $\vec{\beta}$ is $\frac{\pi}{2}$.

Q.3 Answer:

$-2\hat{i} + 6\hat{j} - 10\hat{k}$

Solution:

If $\overrightarrow{\mathrm{F}}$ be the resultant force, then $\overrightarrow{\mathrm{F}} = 2\hat{i} + 4\hat{j} + 2\hat{k}$

Also, $\vec{r} = \overrightarrow{\mathrm{AP}} = -\hat{i} + 3\hat{j} + 2\hat{k}$

$\therefore$ required moment $= \vec{r} \times \overrightarrow{\mathrm{F}} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 3 & 2 \\ 2 & 4 & 2 \end{vmatrix} = -2\hat{i} + 6\hat{j} - 10\hat{k}$

Q.4 Answer:

$\frac{(\hat{\mathrm{i}}+3\hat{\mathrm{j}}-\hat{\mathrm{k}})}{\sqrt{11}}$

Solution:

As given, $\overrightarrow{\mathrm{r}} \times \overrightarrow{\mathrm{a}} = \overrightarrow{\mathrm{b}} \times \overrightarrow{\mathrm{a}} \Rightarrow (\overrightarrow{\mathrm{r}} - \overrightarrow{\mathrm{b}}) \times \overline{\mathrm{a}} = 0$

$\Rightarrow \overrightarrow{\mathrm{r}} - \overrightarrow{\mathrm{b}}$ is parallel to $\overrightarrow{\mathrm{a}} \Rightarrow \overrightarrow{\mathrm{r}} - \overrightarrow{\mathrm{b}} = \mathrm{n}\overrightarrow{\mathrm{a}}, \mathrm{n} \in \mathrm{R}$

$\Rightarrow \overrightarrow{\mathrm{r}} = \overrightarrow{\mathrm{b}} + \mathrm{n}\overrightarrow{\mathrm{a}}$

Similarly, $\overrightarrow{\mathrm{r}} \times \overrightarrow{\mathrm{b}} = \overrightarrow{\mathrm{a}} \times \overrightarrow{\mathrm{b}}$ can be written as $\overrightarrow{\mathrm{r}} = \overrightarrow{\mathrm{a}} + \mathrm{mb}$

where $m \in R$ $\therefore$ From equations (i) and (ii), we get $\mathrm{m} = 1 = \mathrm{n}$ and $\overrightarrow{\mathrm{r}} = \overrightarrow{\mathrm{a}} + \overrightarrow{\mathrm{b}}$

$\Rightarrow \overrightarrow{\mathrm{r}} = \overrightarrow{\mathrm{i}} + 3\overrightarrow{\mathrm{j}} - \overrightarrow{\mathrm{k}}$ and

$|\overrightarrow{\mathbf{r}}| = \sqrt{9+1+1} = \sqrt{11}$

$\therefore \frac{\overrightarrow{\mathrm{r}}}{|\overrightarrow{\mathrm{r}}|} = \frac{\overrightarrow{\mathrm{i}}+3\overrightarrow{\mathrm{j}}-\overrightarrow{\mathrm{k}}}{\sqrt{11}}$

Q.5 Answer:

$-\hat{\mathrm{i}} - 8\hat{\mathrm{j}} + 2\hat{\mathrm{k}}$

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Solution:

Let $\overrightarrow{R} = x\hat{i} + \hat{y} + z\hat{k}$. Then

$\overrightarrow{R} \times \overrightarrow{B} = \overrightarrow{C} \times \overrightarrow{B} \Rightarrow (\overrightarrow{R} - \overrightarrow{C}) \times \overrightarrow{B} = \overrightarrow{0}$

$$\Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x-4 & y+3 & z-7 \\ 1 & 1 & 1 \end{vmatrix} = \overrightarrow{0}$$

$\Rightarrow (y - z + 10)\hat{i} + (z - x - 3)\hat{j} + (x - y - 7)\hat{k} = \overrightarrow{0}$

$\Rightarrow y - z = -10, z - x = 3, x - y = 7$

Also $\overrightarrow{R} \cdot \overrightarrow{A} = 0 \Rightarrow 2x + 0 \cdot y + z = 0 \Rightarrow z = -2x$

Solving, we obtain

$x = -1, y = -8, z = 2$. Hence $\overrightarrow{R} = -\hat{i} - 8\hat{j} + 2\hat{k}$

Q.6 Answer:

60°

Solution:

Let $a = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j} + \hat{k}$

and $b = \frac{1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j} + \hat{k}$

$\therefore \cos\theta = \frac{a \cdot b}{|a||b|}$

$= \frac{\left(\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j} + \hat{k}\right) \cdot \left(\frac{1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j} + \hat{k}\right)}{\sqrt{\frac{1}{2} + \frac{1}{2} + 1}\sqrt{\frac{1}{2} + \frac{1}{2} + 1}}$

$= \frac{1}{2}\left[\frac{1}{2} - \frac{1}{2} + 1\right] = \frac{1}{2} = \cos 60^\circ$

$\therefore \theta = 60^\circ$

Q.7 Answer:

± 4

Solution:

Given, $\overrightarrow{a} \times \overrightarrow{b} = 2\overrightarrow{a} \times \overrightarrow{c} \Rightarrow \overrightarrow{a} \times (\overrightarrow{b} - 2\overrightarrow{c}) = 0$

Since the vectors are non zero.

$\therefore \overrightarrow{b} - 2\overrightarrow{c}$ is parallel to $\overrightarrow{a}$. Let $\overrightarrow{b} - 2\overrightarrow{c} = \lambda\overrightarrow{a}$

Now $|\overrightarrow{b} - 2\overrightarrow{c}|^2 = b^2 + 4c^2 - 4\overrightarrow{b} \cdot \overrightarrow{c}$

$= 16 + 4 - 4 \times 4 \times 1 \times \frac{1}{4} = 16$

i.e., $\lambda^2 a^2 = 16 \Rightarrow \lambda^2 = 16 \Rightarrow \lambda = \pm 4$

Q.8 Answer:

$3\overrightarrow{p}^2$

Solution:

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Suppose $\overrightarrow{p} = p_1\hat{i} + p_2\hat{j} + p_3\hat{k}$

$\overrightarrow{p} \times \hat{i} = p_2\hat{j} \times \hat{i} + p_3\hat{k} \times \hat{i} = -p_2\hat{k} + p_3\hat{j}$

$|\overrightarrow{p} \times \hat{i}|^2 = p_2^2 + p_3^2$

Similarly, $|\overrightarrow{p} \times \hat{j}|^2 = p_3^2 + p_1^2, |\overrightarrow{p} \times \hat{k}|^2 = p_1^2 + p_2^2$

$\therefore \frac{3}{2}\left\{|\overrightarrow{p} \times \hat{i}|^2 + |\overrightarrow{p} \times \hat{j}|^2 + |\overrightarrow{p} \times \hat{k}|^2\right\}$

$= 3\left(p_1^2 + p_2^2 + p_3^2\right) = 3\overrightarrow{p}^2$

Q.9 Answer:

$x = 2, y = -1$

Solution:

Given:

$A = x + 4y\vec{a} + 2x + y + 1\vec{b}$ **and** $B = y - 2x + 2\vec{a} + 2x - 3y - 1\vec{b}$

$3A = 3x + 4y\vec{a} + 32x + y + 1\vec{b}$

$2B = 2y - 2x + 2\vec{a} + 22x - 3y - 1\vec{b}$

In question it is given,

$3A = 2B$

$\Rightarrow 3x + 4y = 2y - 2x + 2 \ldots . 1$

and $32x + y + 1 = 22x - 3y - 1 \ldots . 2$

From the equation 1

$\Rightarrow 7x + 10y = 4$

$\Rightarrow x = \frac{4 - 10y}{7}$

From the equation 2

$2x + 9y = -5$

$\Rightarrow 2\frac{4 - 10y}{7} + 9y = -5$

$\Rightarrow 8 - 20y + 63y = -35$

$\Rightarrow y = -1$

$\Rightarrow x = \frac{4 - 10 - 1}{7} = 2$

Hence, $x = 2, \quad y = -1$**.**

Q.10 Answer:

$\frac{-4\hat{i}+3\hat{j}+\hat{k}}{\sqrt{26}}$

Solution:

$\vec{A} = \hat{i} + \hat{j} + \hat{k}$

$\vec{B} = 2\hat{i} + 3\hat{j} - \hat{k}$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 2 & 3 & -1 \end{vmatrix}$$

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$= \hat{i}(-1-3) - \hat{j}(-1-2) + \hat{k}(3-2)$
$= -4\hat{i} + 3\hat{j} + \hat{k}$

Vector of unit length orthogonal to both the vectors $\vec{A}$ and $\vec{B}$

$= \frac{\vec{A}\times\vec{B}}{|\vec{A}\times\vec{B}|}$
$= \frac{-4i+3j+k}{\sqrt{16+9+1}} = \frac{-4\hat{i}+3\hat{j}+\hat{k}}{\sqrt{26}}$

Q.11 Answer:

$42|\vec{a}\times\vec{b}|^2$

Solution:

$\overrightarrow{BD} = \overrightarrow{a} - 3\overrightarrow{b}, \overrightarrow{AC} = \overrightarrow{a} + 3\overrightarrow{b}$
$\overrightarrow{BD} \times \overrightarrow{AC} = (\overrightarrow{a} - 3\overrightarrow{b}) \times (\overrightarrow{a} + 3\overrightarrow{b}) = 6\overrightarrow{a} \times \overrightarrow{b}$
$\overrightarrow{OD} \times \overrightarrow{OC} = (\overrightarrow{a} - 2\overrightarrow{b}) \times (2\overrightarrow{a} + 3\overrightarrow{b}) = 7\overrightarrow{a} \times \overrightarrow{b}$
$(\overrightarrow{BD} \times \overrightarrow{AC}) \cdot (\overrightarrow{OD} \times \overrightarrow{OC}) = 42(\overrightarrow{a} \times \overrightarrow{b})^2$

Q.12 Answer:

$\sin^{-1}\frac{9}{14}$

Solution:

$\vec{b}\times\vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{vmatrix} = -3\hat{i} - \hat{j} + 2\hat{k}$

If θ is the angle between $\vec{a}$ and the plane containing $\overrightarrow{b}$ and $\overrightarrow{c}$, then

$\cos(90^\circ - \theta) = \left|\frac{\vec{a}\cdot(\vec{b}\times\vec{c})}{|\vec{a}||\vec{b}\times\vec{c}|}\right|$
$= \frac{1}{\sqrt{14}} \cdot \frac{1}{\sqrt{14}}|(-9-2+2)| = \frac{9}{14}$
$\Rightarrow \sin\theta = \frac{9}{14} \Rightarrow \theta = \sin^{-1}\left(\frac{9}{14}\right).$

Q.13 Answer:

Statement Iis true; Statement II is true; Statement II is not the correct explanation of the Statement I

Solution:

Statement II is obviously true.

If we know that A, B and C are collinear, then

$\overrightarrow{AB} = \lambda \overrightarrow{AC}$
$\therefore \overrightarrow{AB} \times \overrightarrow{AC} = 0$
$\Rightarrow \vec{b} - \vec{a} \times \vec{c} - \vec{a} = 0$
$\Rightarrow \vec{b}\times\vec{c} + \vec{a}\times\vec{b} + \vec{c}\times\vec{a} = 0$
i.e., $\vec{a}\times\vec{b} + \vec{b}\times\vec{c} + \vec{c}\times\vec{a} = 0$

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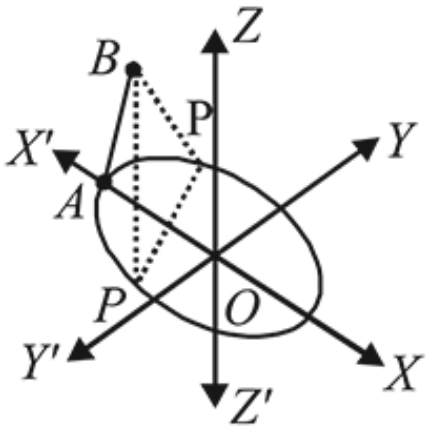
So, it is clear that StatementI is true and Statement II is not the correct explanation of the Statement I

Q.14 Answer:

$\pm\hat{j}$

Solution:

Point P lies on $x^2 + 3y^2 = 3$...(i)



Now from the diagram, according to the given conditions, $AP = AB$ or $(x + \sqrt{3})^2 + (y - 0)^2 = 4$

or $(x + \sqrt{3})^2 + y^2 = 4$..(ii)

Solving (i) and (ii), we get $x = 0$ and $y = \pm 1$ Hence, point P has position vector $\pm\hat{j}$

Q.15 Answer:

$\cos^{-1}\frac{12}{13}$

Solution:

$\left|\vec{P}\right| = 5, \left|\vec{Q}\right| = 12$ **and** $\left|\vec{R}\right| = 13$

From Pythagorean identity we know

$$\cos\theta = \frac{\text{length of adjacent side connected with } \theta}{\text{length of Hypotenuse}}$$

$$\cos\theta = \frac{\vec{Q}}{\vec{R}} = \frac{12}{13} \quad \therefore \quad \theta = \cos^{-1}\frac{12}{13}$$

Q.16 Answer:

$0°$

Solution:

Given: $\vec{P} = \vec{A} + \vec{B}$

Now,

$\vec{P}^2 = \vec{A} + \vec{B}^2$

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$= \vec{A}^2 + \vec{B}^2 + 2\vec{A}\vec{B}\cos\theta$

but $\vec{P} = \vec{A} + \vec{B}$

$\Rightarrow \vec{A} + \vec{B}^2 = \vec{A}^2 + \vec{B}^2 + 2\vec{A}\vec{B}\cos\theta$

$\Rightarrow \vec{A}^2 + \vec{B}^2 + 2\vec{A}\vec{B} = \vec{A}^2 + \vec{B}^2 + 2\vec{A}\vec{B}\cos\theta$

$\Rightarrow \cos\theta = 1 \Rightarrow \theta = 0°$

Q.17 Answer:

6

Solution:

Since, $\vec{a}, \vec{b}$ **and** $\vec{c}$ **are three vectors with magnitude** $|\vec{a}| = |\vec{b}| = 4$ **and** $|\vec{c}| = 2$,

As a **is perpendicular to** $(\vec{b} + \vec{c})$

$\Rightarrow \vec{a} \cdot (\vec{b} + \vec{c}) = 0$ **or** $\vec{a} \cdot \vec{b} + \overrightarrow{a \cdot c} = 0$

$\vec{b}$ **is perpenducular to** $(\vec{c} + \vec{a})$

$\Rightarrow \vec{b} \cdot (\vec{c} + \vec{a}) = 0$ **or** $\vec{b} \cdot \vec{c} + \vec{b} \cdot \vec{a} = 0$

$\Rightarrow \vec{c}$ **is perpendicular to** $(\vec{a} + \vec{b})$

$\Rightarrow \vec{c} \cdot (\vec{a} + \vec{b}) = 0$ **or** $\vec{c} \cdot \vec{a} + \vec{c} \cdot \vec{b} = 0$

From equations (i), (ii) and (iii), we get $\Rightarrow 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0$

Further we know that

$|\vec{a} + \vec{b} + \vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2$

$+\overrightarrow{2a} \cdot \vec{b} + \overrightarrow{2b} \cdot \vec{c} + \overrightarrow{2c} \cdot \vec{a}$

$\Rightarrow |\vec{a} + \vec{b} + \vec{c}|^2 = 4^2 + 4^2 + 2^2 + 0 = 36$

or $|\vec{a} + \vec{b} + \vec{c}| = 6$

Q.18 Answer:

$\frac{7}{3}$

Solution:

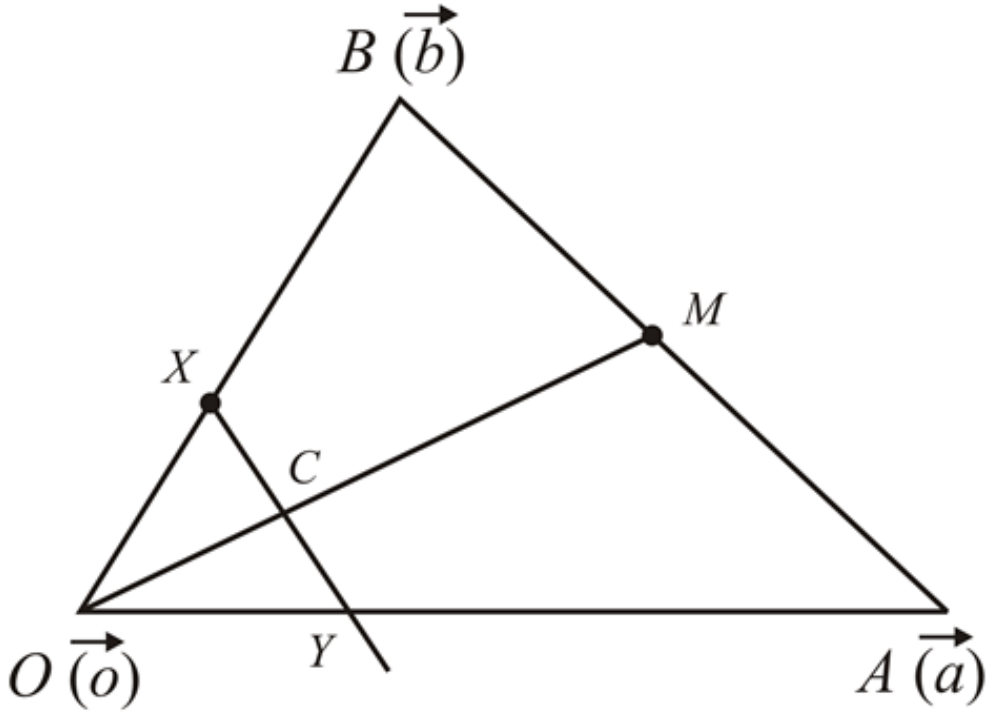
Required value $= \frac{b.a}{|b|} / \frac{a \cdot b}{|a|} = \frac{|a|}{|b|} = \frac{7}{3}.$

Q.19 Answer:

$\frac{2}{7}$

Solution:

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Let O be the origin and the position vectors of A and B be $\vec{a}$ and $\vec{b}$, respectively. Then,

$\overrightarrow{OA} = \vec{a}$ **and** $\overrightarrow{OB} = \vec{b}$

M is the midpoint of AB. Hence,

$OM = \frac{\vec{a}+\vec{b}}{2}$.

Given,

$2\overrightarrow{OC} = \overrightarrow{CM}$

So, C divides OM in $1:2$ ratio.

$\therefore \overrightarrow{OC} = \frac{\vec{a}+\vec{b}}{6}$

Given,

$\overrightarrow{OX} = 2\overrightarrow{XB}$

So, X divides OB in $1:2$ ratio.

$\therefore \overrightarrow{OX} = \frac{2}{3}\overrightarrow{OB}$

$\Rightarrow \overrightarrow{OX} = \frac{2}{3}\vec{b}$

Let $\frac{OY}{YA}$ be equal to k.

So, Y divides OA in $k:1$ ratio.

$\therefore \overrightarrow{OY} = \frac{k}{k+1}\vec{a}$

Now, the points Y, C and X are collinear.

$\therefore \overrightarrow{YC} = l\overrightarrow{CX}$, **where l is a scalar.**

$\Rightarrow \overrightarrow{OC} - \overrightarrow{OY} = l\overrightarrow{OX} - \overrightarrow{OC}$

$\Rightarrow \frac{\vec{a}+\vec{b}}{6} - \frac{k}{k+1}\vec{a} = l\frac{2\vec{b}}{3} - l\frac{\vec{a}+\vec{b}}{6}$.

Comparing the coefficients of $\vec{a}$ and $\vec{b}$ as both are non-collinear vectors, we get,

$\frac{1}{6} - \frac{k}{k+1} = -\frac{l}{6}$ **and** $\frac{1}{6} = \frac{2l}{3} - \frac{l}{6}$.

On solving, we get $l = \frac{1}{3}$ and $k = \frac{2}{7}$.

Substituting the value of k, we get

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$\frac{OY}{YA} = \frac{2}{7}$.

Q.20 Answer:

$\overrightarrow{u}, \overrightarrow{v}$ are orthogonal vectors

Solution:

$\frac{1}{a} = A + (p-1)D; \frac{1}{b} = A + (q-1)D;$
$\frac{1}{c} = A + (r-1)D$
$\therefore \quad q - r = \frac{c-b}{bcD}, r - p = \frac{a-c}{acD}$
$p - q = \frac{b-a}{abD} \Rightarrow \frac{q-r}{a} + \frac{r-p}{b} + \frac{p-q}{c} = 0$
$\Rightarrow \overrightarrow{u} \cdot \overrightarrow{v} = 0$

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